

Self-Improvement Agent Harness: A Deterministic SIA Exemplar

Meta $\rightarrow$ Target $\rightarrow$ Feedback loops with public/private task splits

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1 Abstract

This exemplar documents **template_sia**, a deterministic implementation of the Self-Improvement Agent (SIA) harness contract described in [AI, 2026]. The default pipeline replays fixture-backed generations for the **mini_classify** task; opt-in live mode runs bounded target subprocesses and optional Ollama-backed meta/feedback steps.

Run snapshot. Task **mini_classify**, run 1, 3 generation(s), live=false. Final accuracy=0.8333 over 6 held-out samples. Values are injected by **scripts/z_generate_manuscript_variables.py** after analysis.

Keywords: self-improvement agents, benchmark harness, reproducible evaluation, agent loops

2 Introduction

This exemplar ships **template_sia**, a deterministic research harness for Self-Improvement Agent (SIA) loops [AI, 2026]. It documents how the template repository separates generic orchestration (**infrastructure/sia/**) from a reproducible project surface (**projects/templates/template_sia/**) without vendoring the upstream **upstream SIA orchestrator repository**.

Compared with the AutoResearch exemplar, SIA focuses on **meta** $\rightarrow$ **target** $\rightarrow$ **feedback** generations with public/private task splits rather than candidate-model search and readiness gates. Default CI replays fixture-backed generations; live mode remains opt-in.

Scope. The bundled **mini_classify** task is a threshold classifier on one feature column. Results demonstrate harness wiring and artifact contracts—not state-of-the-art accuracy.

Run contract. Task **mini_classify**, run 1, up to 3 generation(s), live=false.

3 Methodology

3.1 SIA loop

The harness implements a three-agent cycle:

1. **Meta** — proposes or seeds a target agent for generation n .
2. **Target** — runs against public task data and writes `agent_execution.json`.
3. **Feedback** — reads private evaluation metrics and proposes improvements for generation $n + 1$.

SIA loop topology

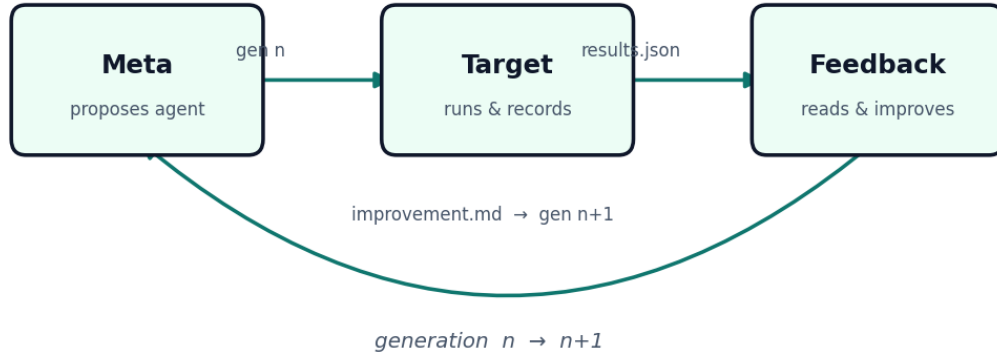


Figure 1: Meta $\rightarrow$ Target $\rightarrow$ Feedback loop topology for the SIA harness, generated programmatically by `write_sia_loop_topology`.

Artifacts land under `output/runs/run_{id}/gen_{n}/` with `target_agent.py`, `agent_execution.json`, optional `improvement.md`, and canonical `results.json`.

3.2 Task layout

Each task exposes:

- `data/public/` — agent-visible instructions and data (`task.md`, `train.csv`, `evaluate.py`).
- `data/private/` — held-out labels for evaluation only.
- `reference/reference_target_agent.py` — deterministic baseline.

The exemplar task `mini_classify` is a threshold classifier on a single feature column.

3.3 Determinism contract

When `sia.live: false` (default), generations replay recorded fixtures from `src/fixtures/recorded_generations/`. CI never executes generated agent code or calls external LLM APIs.

Pass `--live-sia` to `scripts/run_sia_loop.py` for bounded subprocess execution and optional Ollama feedback when a model is configured.

3.4 Per-generation metric overview

The heatmap below provides a compact overview of per-generation accuracy and sample count, enabling at-a-glance comparison across runs.

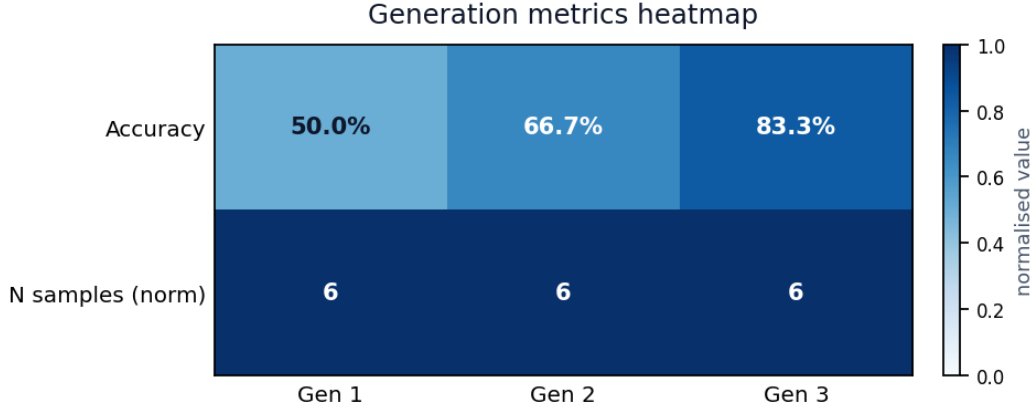


Figure 2: Per-generation metric heatmap showing accuracy and sample count across SIA generations.

4 Results

tbl. 1 summarizes fixture-replay metrics for the bundled run.

Table 1: SIA generation metrics (fixture replay).

Gen	Metric	Value	N
1	accuracy	0.5000	6
2	accuracy	0.6667	6
3	accuracy	0.8333	6

Metric delta (final – first generation): 0.3333.

Final injected token: accuracy=0.8333 (n=6).

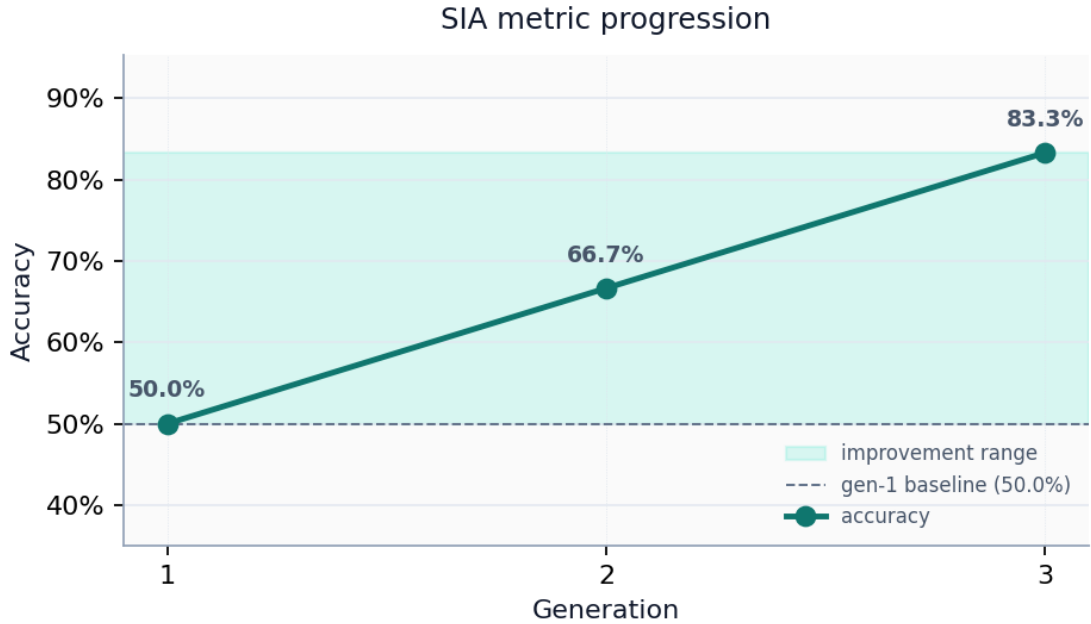


Figure 3: SIA metric progression across generations.

4.1 Incremental improvement

The generation-over-generation accuracy delta quantifies the self-refinement signal at each step of the loop.

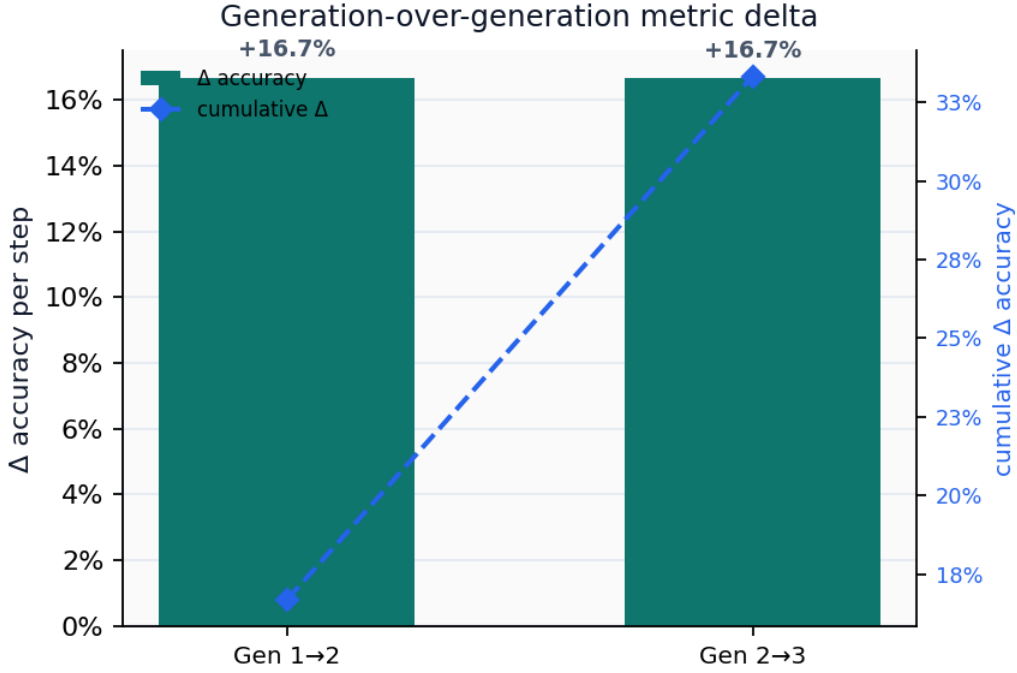


Figure 4: Generation-over-generation metric delta (Δ accuracy) for the SIA loop, illustrating the incremental improvement at each self-refinement step.

5 Conclusion

template_sia demonstrates how to embed the SIA harness contract in the Research Project Template without vendoring upstream orchestration code. Layer 1 (`infrastructure/sia/`) owns task validation, evaluation, context logging, and the generation state machine; Layer 2 wires a minimal classification task, fixture replay, and manuscript tokens.

Non-claims. This tree is a harness and documentation exemplar. Fixture-replay metrics (final accuracy=0.8333) validate wiring only—they are not evidence of state-of-the-art self-improvement. Live self-modification and external LLM calls remain opt-in; default CI never executes generated agent code or claims benchmark parity with [AI, 2026].

6 References

See `references.bib` for BibTeX entries cited in this manuscript, including [AI, 2026] and the template repository DOI from `manuscript/config.yaml`.

References

Hexo AI. Self-improvement agents. 2026. URL <https://arxiv.org/abs/2605.27276>.